BAIO LEARNING SYSTEM • CLASS XI BIOLOGY

Chapter Weightage: 4 Marks

Anatomy of Flowering Plants

Annual Board Examination High-Yield Blueprint & Quick Revision Guide

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1. BLUEPRINT & EXAM PATTERN (4 MARKS TOTAL)

Question Combinations (4 Marks)

- **Pattern A (3M + 1M):** One 3-mark conceptual/ difference question + One 1-mark objective/VSA definition.
- **Pattern B (2M + 2M):** Two 2-mark short reasoning or definition questions.
- **Pattern C (2M + 1M + 1M):** One 2-mark question + Two 1-mark objective questions / MCQs.

Core High-Probability Focus Areas

- **Dicot vs. Monocot Root (3 differences)**
- **Dicot vs. Monocot Stem (3 differences)**
- **Stomatal Apparatus & Bulliform Cells**
- **Radial vs. Conjoint / Open vs. Closed Bundles**
- **Casparian Strips, Stele & Conjunctive Tissue**

2. HIGH-YIELD 3-MARK QUESTIONS

3 MARKS 1. Differentiate between Dicot Root and Monocot Root

Give three important anatomical differences:

- **Xylem Bundles:** 2 to 4 patches (Diarch to Tetrarch) in Dicot root vs. Polyarch (more than 6 patches) in Monocot root.
- **Pith:** Small or inconspicuous in Dicot root vs. Large and well-developed in Monocot root.
- **Secondary Growth:** Present in Dicot root (cambium ring develops) vs. Completely absent in Monocot root.

3 MARKS 2. Differentiate between Dicot Stem and Monocot Stem

- **Vascular Bundles:** Arranged in a ring, open (cambium present) in Dicot stem vs. Scattered throughout ground tissue, closed (cambium absent) in Monocot stem.
- **Hypodermis:** Collenchymatous in Dicot stem vs. Sclerenchymatous in Monocot stem.
- **Ground Tissue & Pith:** Differentiated into cortex, endodermis, pericycle, medullary rays, and pith in Dicot stem vs. Undifferentiated ground tissue with no pith in Monocot stem.

3 MARKS 3. Stomatal Apparatus

- **Stomatal Aperture:** Central pore through which transpiration and gas exchange take place.
- **Guard Cells:** Bean/kidney-shaped in dicots and dumb-bell shaped in grasses; possess chloroplasts; regulate opening and closing of stoma.
- **Subsidiary Cells:** Specialized adjacent epidermal cells enclosing the guard cells that assist in their movement.

3 MARKS 4. Classification of Vascular Bundles

- **Radial:** Xylem and phloem lie on separate radii in an alternating manner (characteristic of roots).
- **Conjoint:** Xylem and phloem are arranged along the same radius (characteristic of stems and leaves).
- **Open vs. Closed:** Open has intrafascicular cambium present between xylem and phloem (capable of secondary growth, e.g., dicot stem); Closed lacks cambium (e.g., monocot stem).

3. HIGH-PRIORITY 2-MARK SHORT ANSWER QUESTIONS

2 MARKS Casparian Strips

A band of suberin (waxy, water-impermeable material) deposited on the radial and tangential walls of root endodermal cells. It regulates water and mineral movement into the stele.

2 MARKS Bulliform Cells

Large, empty, colourless modified epidermal cells along veins in grasses. When turgid, they expose the leaf surface; when flaccid (under water stress), leaves curl inwards to prevent water loss.

2 MARKS Exarch vs. Endarch Xylem

- **Exarch:** Protoxylem lies towards the periphery and metaxylem towards the centre (Roots).
- **Endarch:** Protoxylem lies towards the centre (pith) and metaxylem towards the periphery (Stems).

2 MARKS Trichomes vs. Root Hairs

- **Trichomes:** Multicellular epidermal hairs on stems; prevent transpirational water loss; may be branched/ secretory.
- **Root Hairs:** Unicellular epiblemal elongations; absorb water and minerals from soil.

4. QUICK REVISION COMPARATIVE SUMMARY (TABLE)

Feature	Dicotyledonous Organ	Monocotyledonous Organ
Root Xylem Patches	2 to 4 patches (Diarch to Tetrarch)	More than 6 patches (Polyarch)
Root Pith	Small or inconspicuous	Large and well-developed
Stem Vascular Bundles	Arranged in a ring; Open (cambium present)	Scattered in ground tissue; Closed (cambium absent)
Stem Hypodermis	Collenchymatous	Sclerenchymatous
Stem Phloem Parenchyma	Present	Absent (water-containing cavities present)
Leaf Mesophyll	Differentiated (Palisade + Spongy parenchyma)	Undifferentiated
Guard Cells / Bulliform Cells	Bean-shaped guard cells; Bulliform cells absent	Dumb-bell guard cells (grasses); Bulliform cells present

5. HIGH-FREQUENCY 1-MARK TERMS & DEFINITIONS (VSA / MCQS)

- **Stele:** All tissues inner to the endodermis (Pericycle + Vascular Bundles + Pith).
- **Conjunctive Tissue:** Specialized parenchymatous cells present between xylem and phloem in roots.
- **Starch Sheath:** The innermost cortical layer (endodermis) of dicot stems, rich in starch grains.
- **Medullary Rays:** Radially arranged parenchymatous layers between vascular bundles in dicot stems.

6. MODEL BOARD EXAM QUESTIONS (TARGET: 4/4 MARKS)

1. Mention any three differences between Dicot Stem and Monocot Stem. 3M
2. What are Casparian strips? Mention their location and significance. 2M
3. Describe the structure of stomatal apparatus with a labeled sketch. 3M
4. What are bulliform cells? State their function during water stress in grasses. 2M
5. Distinguish between Radial and Conjoint vascular bundles. 2M
6. Define stele. Name its constituents. 1M
7. Which type of guard cells are found in grasses? 1M